

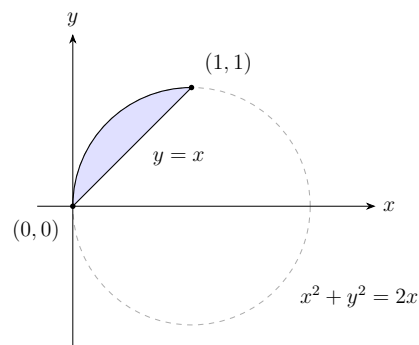
1. **Final answer:** $\boxed{\int_{\pi/4}^{\pi/2} \int_0^{2\cos\theta} r^3 dr d\theta}.$

The Cartesian bounds and the corresponding polar bounds are

$$\begin{cases} 0 \leq x \leq 1, \\ x \leq y \leq \sqrt{2x - x^2} \end{cases} \implies \begin{cases} \pi/4 \leq \theta \leq \pi/2, \\ 0 \leq r \leq 2\cos\theta. \end{cases}$$

The circle is $(x - 1)^2 + y^2 = 1$, or $r = 2\cos\theta$. The retained part lies above $y = x$, so its angles run from $\pi/4$ to $\pi/2$. The boundaries meet at $(0, 0)$ and $(1, 1)$. These polar bounds also give $x = r\cos\theta \leq 2\cos^2\theta \leq 1$.

Since $x^2 + y^2 = r^2$ and $dA = r dr d\theta$, the transformed integrand is r^3 . No evaluation is requested.



Full credit: the single integral in the requested order with the correct region and Jacobian. Accept $r^2 \cdot r$ and angular intervals shifted by $2k\pi$, $k \in \mathbb{Z}$.

2. **Final answer:** $\boxed{M = 4\pi/3}.$

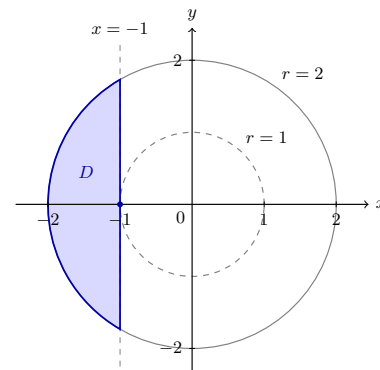
For $-2 \leq x \leq -1$, the condition $x^2 + y^2 \geq 1$ is automatic. The inner circle touches the retained region only at $(-1, 0)$. Thus the Cartesian and polar bounds are

$$\begin{cases} -2 \leq x \leq -1, \\ -\sqrt{4 - x^2} \leq y \leq \sqrt{4 - x^2} \end{cases} \implies \begin{cases} 2\pi/3 \leq \theta \leq 4\pi/3, \\ -\sec\theta \leq r \leq 2. \end{cases}$$

The line $x = -1$ becomes $r\cos\theta = -1$. Since $\cos\theta < 0$ here, dividing $r\cos\theta \leq -1$ reverses the inequality: $r \geq -\sec\theta$. At $r = 2$ the line gives $\cos\theta = -1/2$, determining the angular endpoints. Also, $-\sec\theta \geq 1$ throughout this interval.

Multiplying the density $r^2 - 1$ by the polar area factor r gives

$$M = \int_{2\pi/3}^{4\pi/3} \int_{-\sec\theta}^2 (r^2 - 1)r dr d\theta.$$



Using $\int \sec^4\theta d\theta = \tan\theta + \frac{1}{3}\tan^3\theta + C$,

$$\begin{aligned} M &= \int_{2\pi/3}^{4\pi/3} \left(2 - \frac{1}{4}\sec^4\theta + \frac{1}{2}\sec^2\theta\right) d\theta \\ &= \left[2\theta + \frac{1}{4}\tan\theta - \frac{1}{12}\tan^3\theta\right]_{2\pi/3}^{4\pi/3} = \frac{4\pi}{3}. \end{aligned}$$

At either endpoint, $\tan\theta = \pm\sqrt{3}$, so $\frac{1}{4}\tan\theta - \frac{1}{12}\tan^3\theta = 0$.

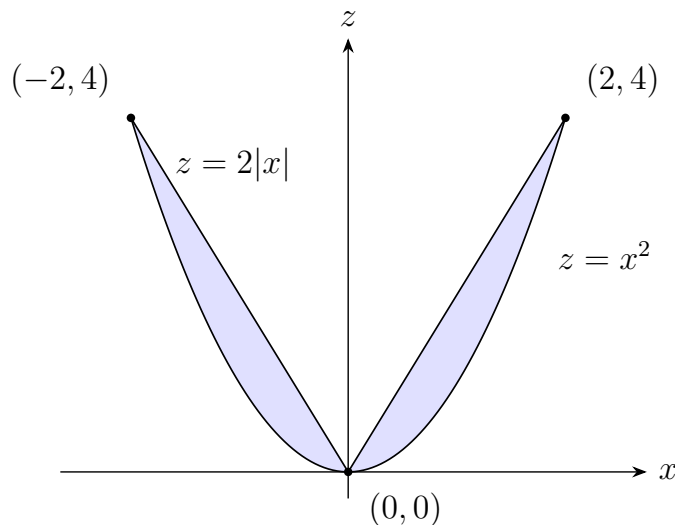
Full credit: any exact expression equal to $4\pi/3$. The question asks for the mass, so an unevaluated integral alone is incomplete.

3. **Final answer:** the two shaded lobes below, with intersections $(x, z) = (-2, 4), (0, 0), (2, 4)$.

The given cylindrical bounds and their restriction to $y = 0$ are

$$\begin{cases} 0 \leq \theta \leq 2\pi, \\ 0 \leq r \leq 2, \\ r^2 \leq z \leq 2r \end{cases} \implies \begin{cases} -2 \leq x \leq 2, \\ x^2 \leq z \leq 2|x|. \end{cases}$$

In Cartesian coordinates the solid satisfies $x^2 + y^2 \leq 4$ and $x^2 + y^2 \leq z \leq 2\sqrt{x^2 + y^2}$. On the entire xz -plane, $r = \sqrt{x^2} = |x|$. Consequently, the section has lower boundary $z = x^2$ and upper boundary $z = 2|x|$. Solving $x^2 = 2|x|$ gives $x = -2, 0, 2$.



Full credit: a correctly shaded sketch of both lobes with all three intersections labeled. Hand-drawn curves need not be to scale. Accept the equivalent three-dimensional labels $(-2, 0, 4)$, $(0, 0, 0)$, and $(2, 0, 4)$, or O for the origin.

4. **Final answer:**

$$\begin{cases} \pi/2 \leq \theta \leq \pi, \\ 0 \leq \varphi \leq \pi, \\ 1 \leq \rho \leq 2. \end{cases}$$

Here φ is measured from the positive z -axis and θ is the azimuthal angle. The conversion is

$$\begin{cases} 1 \leq x^2 + y^2 + z^2 \leq 4, \\ x \leq 0, \\ y \geq 0 \end{cases} \implies \begin{cases} 1 \leq \rho \leq 2, \\ \pi/2 \leq \theta \leq \pi, \\ 0 \leq \varphi \leq \pi. \end{cases}$$

The radii come from $1 \leq \rho^2 \leq 4$. Since $x = \rho \sin \varphi \cos \theta$ and $y = \rho \sin \varphi \sin \theta$, the signs select the second-quadrant azimuthal sector. There is no restriction on z , so both hemispheres are included. This is one quarter of the shell.

Full credit: all three bounds describing this region exactly once, apart from boundary points. Accept any ordering of the inequalities and $\pi/2 + 2k\pi \leq \theta \leq \pi + 2k\pi$ for $k \in \mathbb{Z}$, with the same ρ and φ bounds. No volume calculation is requested.